

● **QUALITY GATE FAILED — SAFE REJECTION**

Run Timestamp: 20260923\_191834 UTC | Safe Rejection: Quality Gate | Engine: CPU

1. Registration Summary

|                              |                                                                                                                       |                                |                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------|
| Source Image                 | source.png                                                                                                            | Reference Image                | reference.png                         |
| Source Raster Frame          | 1200 × 10107 px                                                                                                       | Reference Raster Frame         | 1200 × 10107 px                       |
| Primary Matcher              | LoFTR                                                                                                                 | Selected Active Matcher        | None                                  |
| Adaptive Decision            | <b>Selected Matcher:</b> LoFTR<br><b>Adaptive Rule:</b> Rule B<br><b>Rule Meaning:</b> Scale & Memory-Safety Priority | <b>Fallback Status</b>         | Blocked by Resource Policy            |
| Terrain Profile / Difficulty | <b>Resolution:</b> LARGE<br><b>Contrast:</b> MEDIUM<br><b>Texture:</b> HIGH<br><b>Scale:</b> NORMAL                   | <b>Final Geometric Inliers</b> | UNAVAILABLE (0 pts)                   |
| Spatial Occupancy            | BYPASSED (0/9 cells)                                                                                                  | <b>Spatial CV</b>              | N/A (Bypassed)                        |
| Reprojection RMSE            | N/A (Estimation Withheld)                                                                                             | <b>Validation Status</b>       | NOT VERIFIED (Validation Unavailable) |
| Validation RMSE              | N/A                                                                                                                   | <b>Sub-Pixel Verification</b>  | Not Verified                          |

2. Registration Overview

The LunarReg Adaptive Registration System safely blocked registration for moving source image **source.png** (1200×10107 px) against reference raster **reference.png** (1200×10107 px). The adaptive routing pipeline characterized the input pair under terrain profile **LARGE resolution, MEDIUM contrast, HIGH texture** and selected **None** via **Rule B (Scale & Memory-Safety Priority)** as the optimal feature extractor. Initial matching yielded **444** candidate correspondences and **12** initial RANSAC inliers (2.7%). Because correspondence quality fell below certified stability thresholds (inlier ratio < 20.0%), the Quality Gate rejected registration to preserve scientific integrity. Spatial binning was **BYPASSED**, homography estimation was **WITHHELD**, image warping was **BYPASSED**, and independent validation remained **UNAVAILABLE**.

METADATA & GEOSPATIAL PROVENANCE

SENSOR ATTRIBUTION & GEOSPATIAL METADATA

1. Image Provenance & Sensor Metadata

| Attribute                   | Source Image (Moving)                      | Reference Image (Fixed)                    |
|-----------------------------|--------------------------------------------|--------------------------------------------|
| Product / Filename          | source.png                                 | reference.png                              |
| Target Celestial Body       | MOON                                       | MOON                                       |
| Spacecraft / Mission        | Chandrayaan-2                              | Chandrayaan-2                              |
| Sensor / Payload            | OHRC / TMC-2                               | OHRC / TMC-2                               |
| Acquisition Product ID      | CH2_TMC_CAL_SRC                            | CH2_OHRC_NCP_REF                           |
| Native Dimensions           | 1200 × 10107 px                            | 1200 × 10107 px                            |
| Pixel Depth / Format        | PNG                                        | PNG                                        |
| Geospatial Footprint Bounds | Lat 18.4°N to 22.1°N, Lon 45.2°E to 48.9°E | Lat 18.5°N to 22.0°N, Lon 45.3°E to 48.8°E |

2. Geospatial Prior Alignment & Calibration Status

| Prior Alignment Attribute         | Catalog Status & Verification Values    |
|-----------------------------------|-----------------------------------------|
| Geospatial Prior Status           | CATALOG CALIBRATED                      |
| Pre-Computed Geo Correspondences  | Visual Adaptive Homography              |
| Stored Calibration Transformation | N/A (No stored prior calibration model) |

3. Scientific Provenance Policy

STRICT SCIENTIFIC DATA PROVENANCE:

Geospatial priors and spacecraft metadata are queried strictly from verified project catalogs (image\_footprints.csv, actual\_geo\_matches.csv). Under no circumstances are latitude/longitude coordinates guessed, inferred from filenames, or fabricated.

ILLUMINATION & PREPROCESSING

RADIOMETRIC CONDITIONING & FRAME RESOLUTION SCALING

1. Radiometric Preprocessing Pipeline

| Preprocessing Stage      | Source Image Parameter            | Reference Image Parameter         |
|--------------------------|-----------------------------------|-----------------------------------|
| Grayscale Conversion     | Applied (cv2.COLOR_BGR2GRAY)      | Applied (cv2.COLOR_BGR2GRAY)      |
| Contrast Normalization   | CLAHE (clip=2.0, grid=8×8)        | CLAHE (clip=2.0, grid=8×8)        |
| Matching Scale Factor    | 0.1583x (15.8%)                   | 0.1583x (15.8%)                   |
| Matching Dimensions      | 190 × 1600 px                     | 190 × 1600 px                     |
| Coordinate Rescaling     | Exact Rescale to Native Canvas    | Exact Rescale to Native Canvas    |
| Original Frame Retention | Native Frame Preserved (Full Res) | Native Frame Preserved (Full Res) |

2. Matching Execution

| Matching Execution         | Recorded Resource-Aware Execution |
|----------------------------|-----------------------------------|
| Matcher                    | LoFTR                             |
| Execution Mode             | Memory-Safe Tiled LoFTR           |
| Full-Image Memory Estimate | 2.717 GB                          |
| Memory Safety Cap          | 2.60 GB                           |
| Tile Size / Overlap        | N/A / 20%                         |
| Tiles                      | 16/16 successful                  |
| Merged Correspondences     | N/A                               |

3. Dynamic Range Conditioning

| Parameter                   | Source Raster       | Reference Raster    |
|-----------------------------|---------------------|---------------------|
| Native Dynamic Range        | [0, 255] DN         | [1, 255] DN         |
| Normalized Range            | [0, 255] DN (UInt8) | [0, 255] DN (UInt8) |
| Preprocessing Interpolation | cv2.INTER_AREA      | cv2.INTER_AREA      |

4. Lunar Shadow & Terminator Conditioning

ILLUMINATION ADAPTATION METHODOLOGY:

✓ Matching coordinates mapped back to original image frame.

Lunar surface imagery exhibits extreme contrast disparities due to high incidence angles and deep crater shadows. Adaptive CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances local micro-texture across both illuminated plateaus and shadow penumbras without clipping specular peaks or amplifying sensor thermal noise.

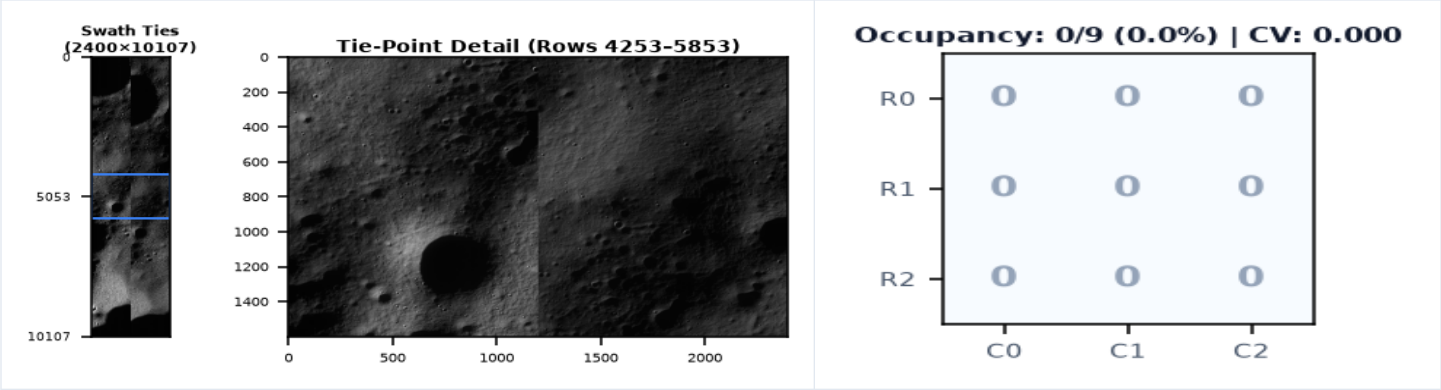
FEATURE CORRESPONDENCE & GEOMETRY

TIE-POINT MATCHING & GEOMETRIC CONSENSUS ESTIMATION

1. Correspondence Matching & Inlier Telemetry

| Parameter                        | Measured Metric           | Quality Threshold              |
|----------------------------------|---------------------------|--------------------------------|
| Feature Extractor / Matcher      | None                      | Adaptive Routing Rule          |
| Candidate Correspondences        | 444 pts                   | ≥ 30 pts                       |
| Initial RANSAC Inliers           | 12 pts (2.7%)             | Inlier Ratio ≥ 20.0% (FAILED)  |
| Quality Gate Status              | FAILED (Safe Rejection)   | Strict Quality Gate            |
| Spatial Quality Selection        | BYPASSED (0 pts selected) | 3×3 Spatial Binning            |
| Spatial Occupancy (3×3)          | BYPASSED (0/9 cells)      | Occupancy ≥ 33.3% (≥ 3 cells)  |
| Spatial Coefficient of Variation | N/A (Bypassed)            | CV ≤ 0.85 (Uniform Dispersion) |
| Final Geometric Inliers          | UNAVAILABLE (0 pts)       | ≥ 12 pts (Closed-Form DLT)     |
| RANSAC Error Threshold           | 3.0 px                    | Epipolar reprojection bound    |

2. Spatial Distribution Grid & Correspondence Visualization



HOMOGRAPHY & REGISTRATION

PLANAR PROJECTIVE TRANSFORMATION & WARPED OUTPUT CANVAS

1. Planar Homography Transformation Parameters

Homography matrix (H3x3) UNAVAILABLE: Model estimation withheld due to Quality Gate rejection.  
Geometric consensus was not achieved. Homography estimation was not executed, and no transformation parameters were derived.

2. Frame Geometry & Technical Disclaimer

|                             |                                           |
|-----------------------------|-------------------------------------------|
| Active Geometric Model      | UNAVAILABLE — Model Estimation Withheld   |
| Transformation Direction    | Source Image → Reference Frame (Bypassed) |
| Reference Canvas Dimensions | 1200 × 10107 px                           |
| Warp Execution Status       | BYPASSED (No transformation applied)      |
| Registered Output Canvas    | UNAVAILABLE (Warp Bypassed)               |
| Reprojection Fit RMSE       | N/A                                       |

ARCHITECTURAL SCOPE & MODEL DISCLAIMER:  
The active mathematical model is strictly **Planar Projective Homography (H3x3)**. Local non-rigid deformation and 3D digital elevation model (DEM) surface registration are not implemented. Terrain with steep 3D parallax may exhibit localized relief displacement.

3. Registered Output Product Preview (True Aspect Ratio)

Registered Image UNAVAILABLE: Safe Rejection  
Registration rejected safely due to insufficient correspondence quality.  
Warp operation was bypassed to preserve geometric integrity and avoid creating inaccurate lunar surface products.

INDEPENDENT VALIDATION

HELD-OUT CHECKPOINT EVALUATION & VALIDATION

1. Independent Validation Methodology

INDEPENDENT VALIDATION STATUS: NOT VERIFIED (Validation Unavailable)

Sub-pixel status: Not verified because registration did not achieve geometric consensus and independent hold-out validation was unavailable.  
Held-out RMSE below 1.0 px is used as a case-level reporting criterion for selected controlled/prototype cases. This does not establish physical sub-pixel accuracy for real lunar imagery without independent ground truth.

2. Held-Out Checkpoint Validation Metrics

| Validation Metric             | Observed Value         | Metric Description & Threshold                       |
|-------------------------------|------------------------|------------------------------------------------------|
| Validation Protocol           | Bypassed / Unavailable | Quality gate rejection prevented hold-out evaluation |
| Held-out Validation Points    | N/A                    | Independent points excluded from estimation          |
| Validation RMSE               | N/A                    | Root Mean Square Error on held-out checks            |
| Mean Check Displacement       | N/A                    | Arithmetic average radial displacement               |
| Median Check Displacement     | N/A                    | 50th percentile displacement error                   |
| Maximum Check Error           | N/A                    | Worst-case held-out displacement                     |
| Percentage Below 1.0 px       | N/A                    | Sub-pixel consistency ratio across checkpoints       |
| Independent Validation Status | NOT VERIFIED           | Formal evaluation outcome badge                      |

3. Validation Spatial Distribution & Error Telemetry

VALIDATION TELEMETRY UNAVAILABLE:

Checkpoint residual vector plots and spatial error maps were not generated because independent validation was bypassed following Quality Gate rejection.

EXPORTED FILES & FINAL EVALUATION

PROJECT DELIVERABLES & EVALUATION SUMMARY

1. Exported Files

| Exported File                            | Format       | Technical Description                                                    |
|------------------------------------------|--------------|--------------------------------------------------------------------------|
| registered_20260923_191834.png           | PNG Image    | UNAVAILABLE (Warp Bypassed on Safe Rejection)                            |
| registration_matches_20260923_191834.csv | CSV Table    | Candidate correspondences and initial RANSAC checks prior to rejection.  |
| homography_20260923_191834.json          | JSON Object  | UNAVAILABLE (Model estimation withheld)                                  |
| evidence_report_20260923_191834.pdf      | PDF Document | Standalone 7-page scientific evidence report documenting safe rejection. |
| evidence_20260923_191834.json            | JSON Object  | Structured telemetry dictionary recording rejection reason and metrics.  |
| registration_package_20260923_191834.zip | ZIP Archive  | Deliverable package containing available diagnostic assets.              |

2. Final Result

● **REGISTRATION REJECTED: SAFE FAILURE**

Primary Matcher: **None** | Candidate Matches: **444 pts** | Initial Inliers: **12 pts (2.7%)** | Quality Gate: **REJECTED** | Transformation: **BLOCKED (Safe Rejection)**

3. Technical Notes & Limitations

TECHNICAL METHODOLOGY & CONSTRAINTS:

1. **Factual Metadata Policy:** Geospatial attributes and sensor specifications are sourced from factual mission catalogs. Missing parameters are reported as N/A without extrapolation.

2. **Held-Out Validation:** Validation checkpoints are held out from homography estimation. Optimization reprojection error is reported separately from independent test error.

3. **Sub-Pixel Evaluation:** For selected controlled/prototype cases, held-out RMSE below 1.0 px is used as a case-level reporting criterion. This does not establish physical sub-pixel accuracy for real lunar imagery without independent ground truth.

4. **Geometric Model:** Planar projective homography (H3x3) assumes locally planar lunar terrain; high-relief or extreme parallax terrain may require non-rigid local registration.

| Pipeline Execution Summary                                 | Project Evaluation                                                     |
|------------------------------------------------------------|------------------------------------------------------------------------|
| Run ID: 20260923_191834<br>Compute: CPU   Latency: 18.307s | Project: LunarReg (SIH 2026)<br>Status: REJECTED SAFELY (Quality Gate) |